

Biztel.ai - Python Assignment

Objective:

Write a python code which takes user input as Input mentioned in the document and outputs the output mentioned in the document

Assignment details:

1. There exists a rectangular document with non-overlapping words in it. A document can be considered as a 2D plane with top left as the origin. You have been given a set of words with scaled rectangular bounding box containing that word along with the document size. These boxes are scaled and stored in a range of 0 to 1 in a text file **words_position_dict.txt**.

Rectangular bounding boxes consist of information representing, bounding box width, bounding box height, bounding box left and bounding box top, in the same sequence. Bounding box left represent distance from the top left corner of the box to left margin of the document and bounding box top represents distance of the top left corner of the box to the top edge of the document.

Given that a user can draw a random rectangular bounding box over the actual document whose coordinates are given as [x1, y1, x3, y3], you have to find the words that would be a part of this user made bounding box.

Note : x1,y1 are the actual coordinates of top left of the box and x3, y3 are the actual coordinates for the lower right of the bounding box.

Input

1. **words_position_dict.txt**
Dictionary of words and
2. Width = 1263, height = 1644
3. User drawn bounding box = [358,140,498,171]

Output:

1. Word / Words enclosed within the bounding box